CyberShield CLI

Cybersecurity Incident Tracker

Prajakta Roy Arpa(25-62049-2) and Sarfia Sayda Shamanta(25-62726-2)

Submission date: 28 March, 2026

1. About the Project

CyberShield CLI is a terminal-based cybersecurity incident tracker built in Python. It lets users log, manage, and report on security incidents directly from the command line. The interface is powered by the rich library for clean, colour-coded output. All data is saved to a JSON file so it persists between sessions.

2. Features

Create Incident

This method adds a new incident by asking the user for a title, attack type, and severity level. A date and unique ID is assigned automatically. Input is validated and empty fields and invalid severity values are rejected.

List Incidents

With this method all saved incidents are in a formatted table with columns for ID, Title, Attack Type, Severity, Status, and Date. Shows a friendly message if no incidents are existing yet.

Update Incident Status

This method allows the user to change the status of an existing incident by entering its ID. Valid status options are Open, Resolved, and Closed. Invalid IDs and non-numeric input are handled automatically.

Delete Incident

This method removes an incident by ID. The user is asked to confirm before deletion so records are not accidentally lost.

Search Incidents

This method filters the incident list by keyword, severity level. The results are displayed in the same formatted table. If nothing matches a clear message is shown.

Severity Report

This method generates a summary table showing how many incidents exist at each severity level along with a total count. It is useful for a quick overview of the current security situation.

Auto-Save & Persistence

All the data is automatically saved to incidents.json after every create, update, or delete action. On startup the program loads existing data from the file, so nothing gets lost between sessions.

3. Setup & Run Commands

Step 1 : Creating a virtual environment

```
python -m venv .venv
```

Step 2 : Activating the virtual environment

```
.venv\Scripts\activate
```

Step 3 : Installing required packages

```
pip install -r requirements.txt
pip install rich
```

Step 4 : Run the application

```
python main.py
```

Step 5 : Freeze dependencies

```
pip freeze > requirements.txt
```

4. Screenshots

```
PS C:\Users\windows\OneDrive\Documents\GitHub\CybersecurityTracker> & c:\Users\windows\OneDrive\Documents\GitHub\CybersecurityTracker\venv\Scripts\Activate.ps1
(venv) PS C:\Users\windows\OneDrive\Documents\GitHub\CybersecurityTracker> & c:\Users\windows\OneDrive\Documents\GitHub\CybersecurityTracker\venv\Scripts\python.exe
 c:/Users/windows/OneDrive/Documents/GitHub/CybersecurityTracker/main.py

                    CyberShield CLI

──────────────────── CYBERSECURITY INCIDENT TRACKER ────────────────────
1. Create Incident
2. List Incidents
3. Update Incident Status
4. Delete Incident
5. Search Incidents
6. Severity Report
7. Exit

Enter your choice:
```

```
(venv) PS C:\Users\windows\OneDrive\Documents\GitHub\CybersecurityTracker> & c:\Users\windows\OneDrive\Documents\GitHub\CybersecurityTracker\venv\Scripts\python.exe
 c:/Users/windows/OneDrive/Documents/GitHub/CybersecurityTracker/main.py
6. Severity Report
7. Exit

Enter your choice: 1
──────────────────────────────── New Incident ────────────────────────────
Enter incident title: 1
Enter attack type: Network
Enter severity (Low/Medium/High): High
 Incident #3 created successfully!
──────────────────── CYBERSECURITY INCIDENT TRACKER ────────────────────
1. Create Incident
2. List Incidents
3. Update Incident Status
4. Delete Incident
5. Search Incidents
6. Severity Report
7. Exit

Enter your choice:
```

```
Enter your choice: 2
```

Cybersecurity Incidents

ID	Title	Attack Type	Severity	Status	Date
1	Test 1	Network	High	Open	2026-03-28 08:40
2	Test 2	Firewall	Medium	Open	2026-03-28 08:40
3	1	Network	High	Open	2026-03-28 09:15

```
CYBERSECURITY INCIDENT TRACKER
1. Create Incident
2. List Incidents
3. Update Incident Status
4. Delete Incident
5. Search Incidents
6. Severity Report
7. Exit
```

```
c:/Users/windows/OneDrive/Documents/GitHub/CybersecurityTracker/main.py
5. Search Incidents
6. Severity Report
7. Exit

Enter your choice: 3
Update Status
Enter incident ID: 1
Enter new status (Open/Resolved/Closed): Closed
Incident updated successfully!
CYBERSECURITY INCIDENT TRACKER
1. Create Incident
2. List Incidents
3. Update Incident Status
4. Delete Incident
5. Search Incidents
6. Severity Report
7. Exit

Enter your choice:
```

```
(venv) PS C:\Users\windows\OneDrive\Documents\GitHub\CybersecurityTracker> & c:\Us
ers\windows\OneDrive\Documents\GitHub\CybersecurityTracker\venv\Scripts\python.exe
 c:/Users/windows/OneDrive/Documents/GitHub/CybersecurityTracker/main.py
5. Search Incidents
6. Severity Report
7. Exit

Enter your choice: 4
──────────────────────────────── Delete Incident ────────────────────────────────
Enter incident ID: 1
Are you sure you want to delete? (yes/no): yes
Incident deleted successfully!
──────────────────────────── CYBERSECURITY INCIDENT TRACKER ────────────────────────────
1. Create Incident
2. List Incidents
3. Update Incident Status
4. Delete Incident
5. Search Incidents
6. Severity Report
7. Exit

Enter your choice:
```

```
ers\windows\OneDrive\Documents\GitHub\CybersecurityTracker\venv\Scripts\python.exe
 c:/Users/windows/OneDrive/Documents/GitHub/CybersecurityTracker/main.py
Enter your choice: 5
──────────────────────────────── Search Incidents ────────────────────────────────
Keyword to search (or leave blank): Test 2
Filter by severity (Low/Medium/High or blank):
                                Search Results
```

ID	Title	Attack Type	Severity	Status	Date
2	Test 2	Firewall	Medium	Open	2026-03-28 08:40

```
──────────────────────────── CYBERSECURITY INCIDENT TRACKER ────────────────────────────
1. Create Incident
2. List Incidents
3. Update Incident Status
4. Delete Incident
5. Search Incidents
6. Severity Report
7. Exit
```

```
ers\windows\OneDrive\Documents\GitHub\CybersecurityTracker\venv\Scripts\python.exe
 c:/Users/windows/OneDrive/Documents/GitHub/CybersecurityTracker/main.py
```

Incident Severity Report

Severity	Count
High	1
Medium	1
Low	0
TOTAL	2

CYBERSECURITY INCIDENT TRACKER

```
1. Create Incident
2. List Incidents
3. Update Incident Status
4. Delete Incident
5. Search Incidents
6. Severity Report
7. Exit

Enter your choice:
```

```
1. Create Incident
2. List Incidents
3. Update Incident Status
4. Delete Incident
5. Search Incidents
6. Severity Report
7. Exit

Enter your choice: 7

Data has been saved. Stay secured and goodbye!

(venv) PS C:\Users\windows\OneDrive\Documents\GitHub\CybersecurityTracker>
```

date is ready, click to restart. Ln 97, Col 26 Spaces: 4 UTF-8 CRLF { } Python venv (3.14.